# Numerical Linear Algebra for Computational Science and Information Engineering

## Lecture 17
## Introduction to Communication-Avoiding Algorithms

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# Outline


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# Introduction

CS294/Math270 – Demmel and Grigori (2016)
MIT 6.172 – Shun (2018)

# Cost of algorithm deployment

- Deploying an algorithm has **two costs** measured in **time** (or **energy**):
  - **Arithmetic** (floating-point operations, FLOPs)

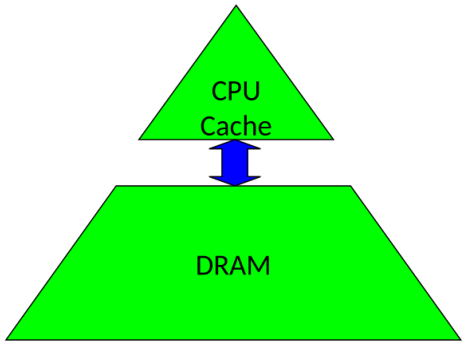
$$\boxed{\#\text{ of }\textbf{FLOPs} \div (\#\text{ of }\textbf{FLOPs per cycle} \times \#\textbf{ cycles per unit of time})}$$

  - **Communication** due to data movement between
    - **levels of a memory hierarchy**:

      **sequential** part of a program

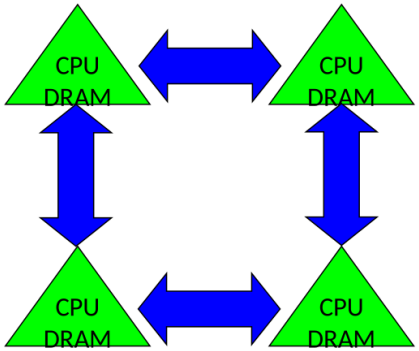
      transfers between DRAM, L3 cache, L2 cache, L1 cache and registers

      cache hit, cache miss, ...




    - **processors over a network**:

      **parallel** part of a program




$$\boxed{\#\text{ of }\textbf{messages} \times \textbf{latency} + \#\text{ of }\textbf{words} \div \textbf{bandwidth}}$$

$$\boxed{\textbf{idle}\text{ time}}$$

# Roofline model

- **Algorithm** on given **hardware** characterized by **arithmetic intensity**:

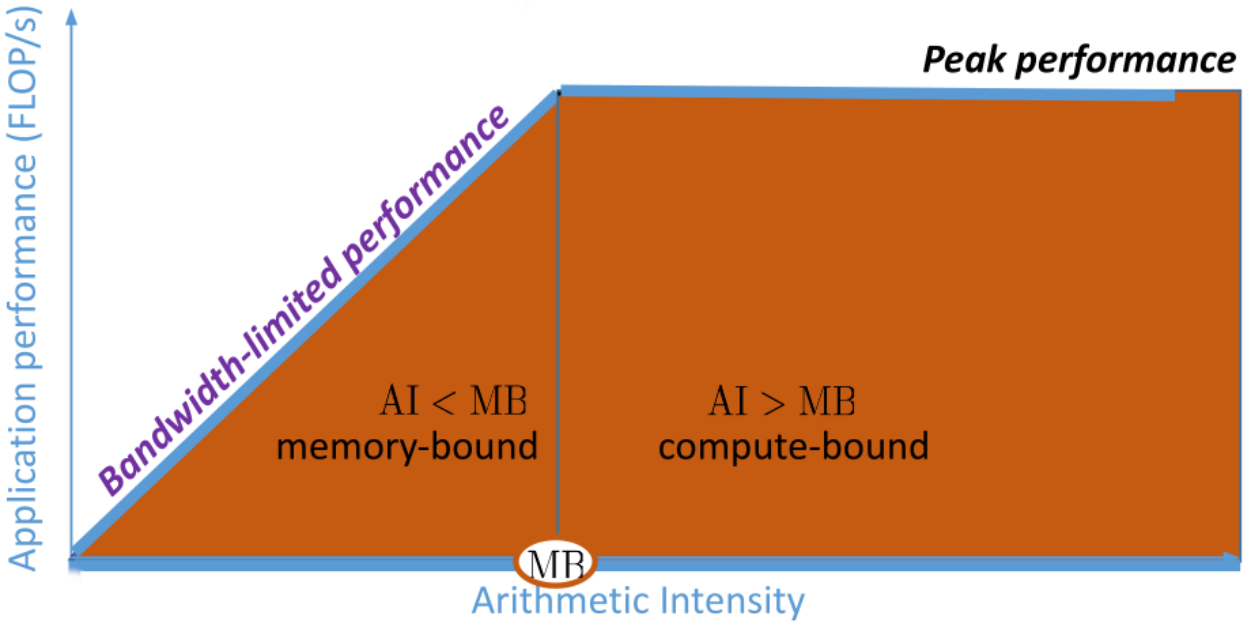
## Arithmetic intensity (AI)

Ratio of number of FLOPs over amount of data moved.

- **Hardware** characterized by **machine balance**:

## Machine balance (MB)

Ratio of peak floating-point performance over peak memory bandwidth.

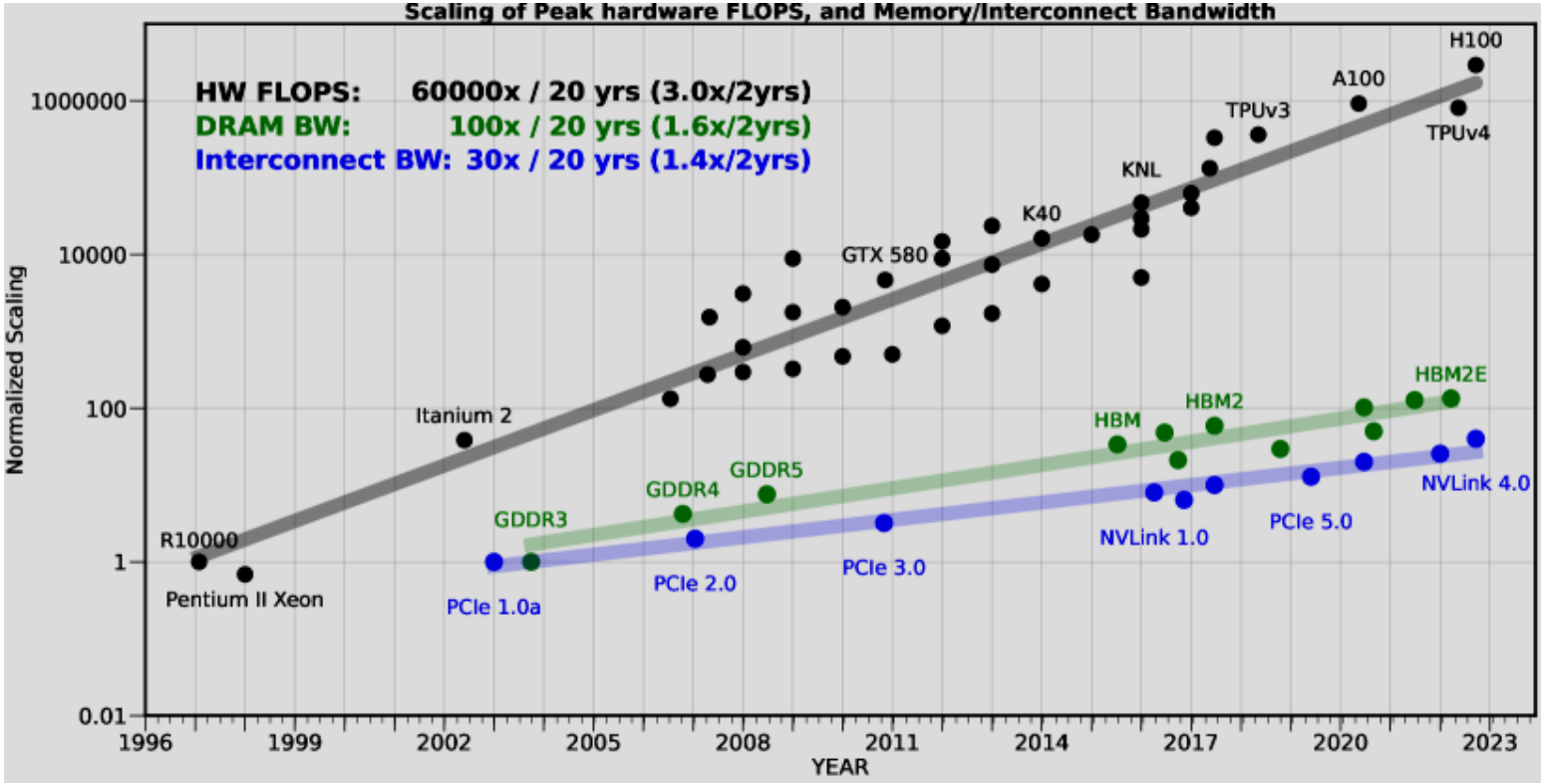


# Memory wall

- Cost to move data much greater than cost of arithmetic:

$$\textbf{time per FLOP} \ll \frac{1}{\textbf{bandwidth}} \ll \textbf{latency}$$

- Peak floating-point performance evolves faster than memory bandwidth:

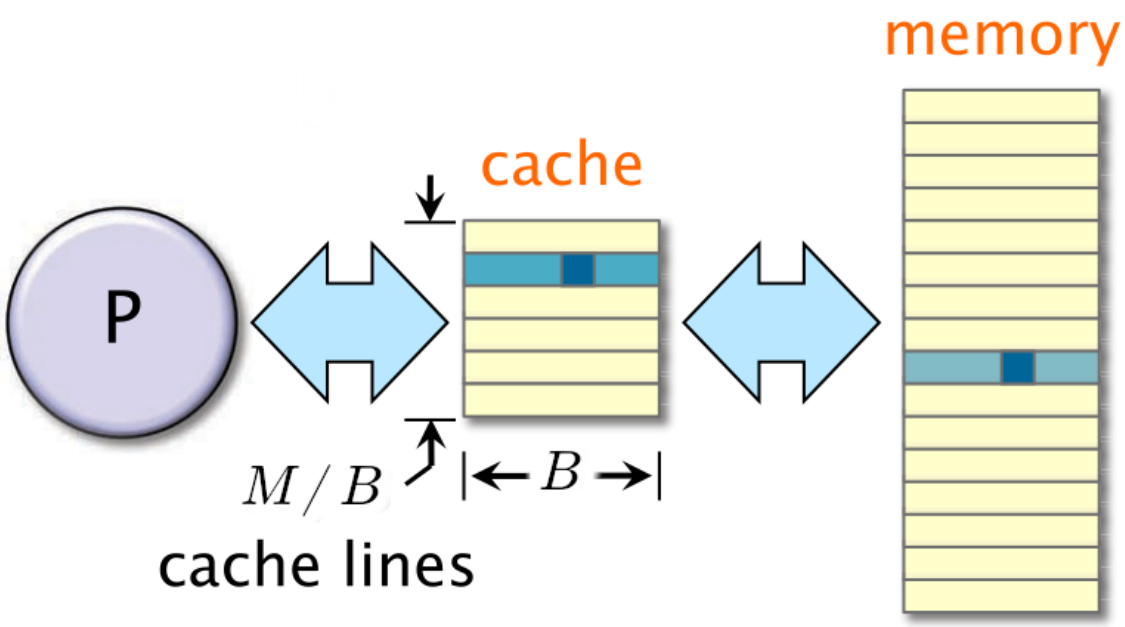



- Relative cost of algorithms due to communication larger every year

Gholami, A., Yao, Z., Kim, S., Hooper, C., Mahoney, M.W., Keutzer, K. (2024). AI and Memory Wall. arXiv:2403.14123v1.

# Ideal cache model

- An **ideal cache model** is introduced for the analysis of algorithms:




- Model parameters:
  - Two-level hierarchy
  - Cache size of $M$ bytes
  - Cache line length of $B$ bytes
  - Fully associative cache (cache lines can be stored anywhere in cache)
  - Least-recently used (LRU) cache replacement policy
  - Arbitrary large main memory

# Tall caches

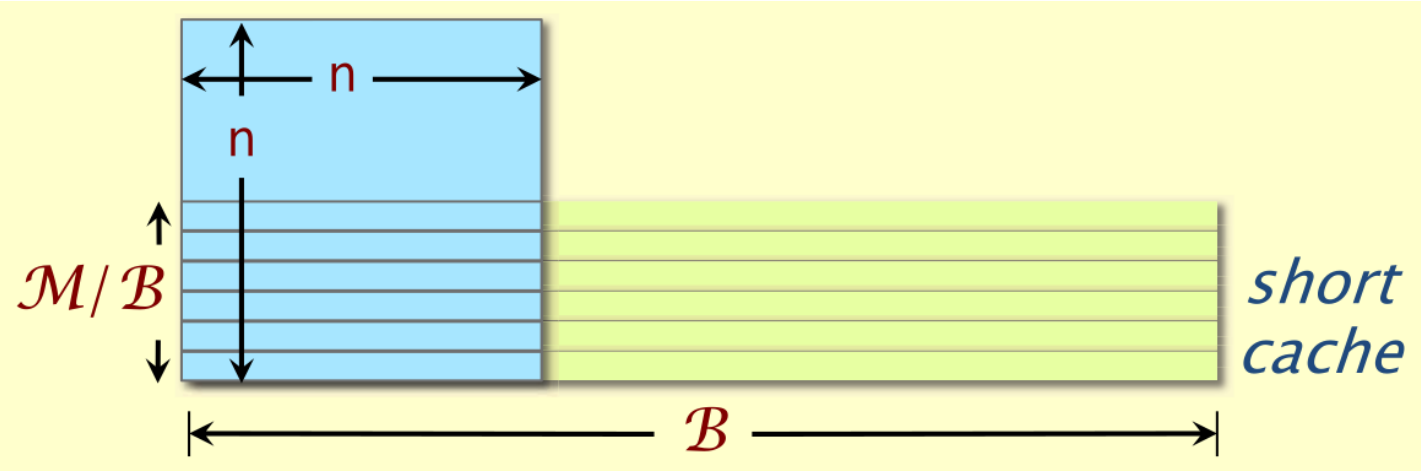
- Several of the coming cache analyses assume a tall cache:

## Tall cache assumption

A cache of size $M$ bytes with $N/B$ lines of $B$ bytes is **tall** if $B^2 < cM$ for some sufficiently small $c < 1$.

Problem of **short cache**:

- a $n \times n$ **sub**matrix may not fit in cache, even if $n^2 < cM$




i.e., cache lines store only contiguous data, and consecutive rows of a row major submatrix are not contiguous in memory.

# Communication lower bounds

- Assuming a fast memory of size $M$, Ballard et al. (2011) derive **lower bounds** on communication for valid all methods of **direct linear algebra**, **dense** or **sparse**:

$$\textbf{bandwidth cost}: \ \# \text{ of words moved} = \Omega\left(\frac{\# \text{ of arithmetic operations}}{\sqrt{M}}\right)$$

$$\textbf{latency cost}: \ \# \text{ of messages sent} = \Omega\left(\frac{\# \text{ of arithmetic operations}}{\sqrt{M^3}}\right)$$

- These bounds hold for
  - Matrix-multiply, LU, QR, eigensolving, SVD, tensor contraction, ...
  - Some whole programs (sequences of these operations, no matter how individual operations are interleaved, e.g. $A^k$)
  - **Sequential** and **parallel** algorithms
  - Some graph-theoretic algorithms (e.g. Floyd-Warshall)
- Significant efforts by the NLA community to rethink algorithms to try and achieve these bounds.

Ballard, G., Demmel, J., Holtz, O., & Schwartz, O. (2011). Minimizing communication in numerical linear algebra. SIAM Journal on Matrix Analysis and Applications, 32(3), 866-901.

# Matrix-matrix multiplication

MIT 6.172 – Shun (2018)

# Cache misses analysis

- Matrix-matrix multiplication with row major data:

```
for i = 0, ..., n-1                                  //zero-based indexing
  for j = 0, ..., n-1
    C[i*n+j] := 0
    for k = 0, ..., n-1
      C[i*n+j] := C[i*n+j] + A[i*n+k] * B[k*n+j]
```

Cache miss complexity, $Q(n)$, **dominated by access to** $B$.

Assume a **tall** and **ideal cache**. There are 3 cases:

- Case 1: $n > cM/B$

  $B$ misses on every access $\implies Q(n) = \Theta(n^3)$
- Case 2: $c'M^{1/2} < n < cM/B$

  $B$ exploits spatial locality $\implies Q(n) = \Theta(n^3/B)$
- Case 3: $n < cM^{1/2}$

  $B$ fits in cache $\implies Q(n) = \Theta(n^2/B)$

# Blocked (tiled) matrix multiplication

- Introduce a block size $s$:

```
for i_1 = 0, s, ..., n − s                                  //zero-based indexing
  for j_1 = 0, s, ..., n − s
    for k_1 = 0, s, ..., n − s
      for i = i_1, ..., i_1 + s − 1
        for j = j_1, ..., j_1 + s − 1
          C[i * n + j] := 0
          for k = k_1, ..., k_1 + s − 1
            C[i * n + j] := C[i * n + j] + A[i * n + k] * B[k * n + j]
```

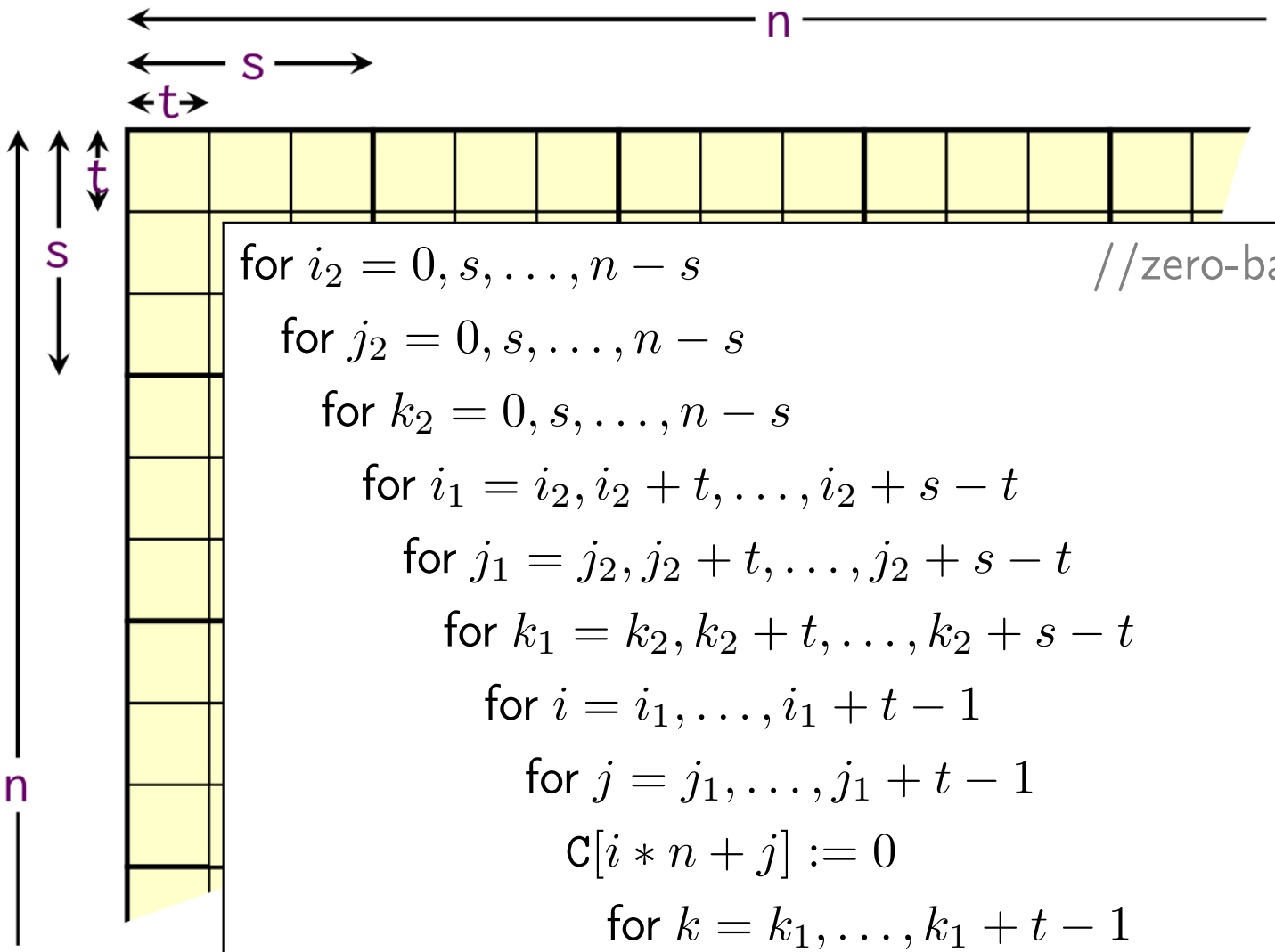
**Tune** $s$ so that the submatrices fit in cache, i.e., $s = \Theta(M^{1/2})$.

$\implies$ **cache-aware** algorithm, **architecture dependent**.

Then $\Theta(s^2/B)$ misses per submatrix so that

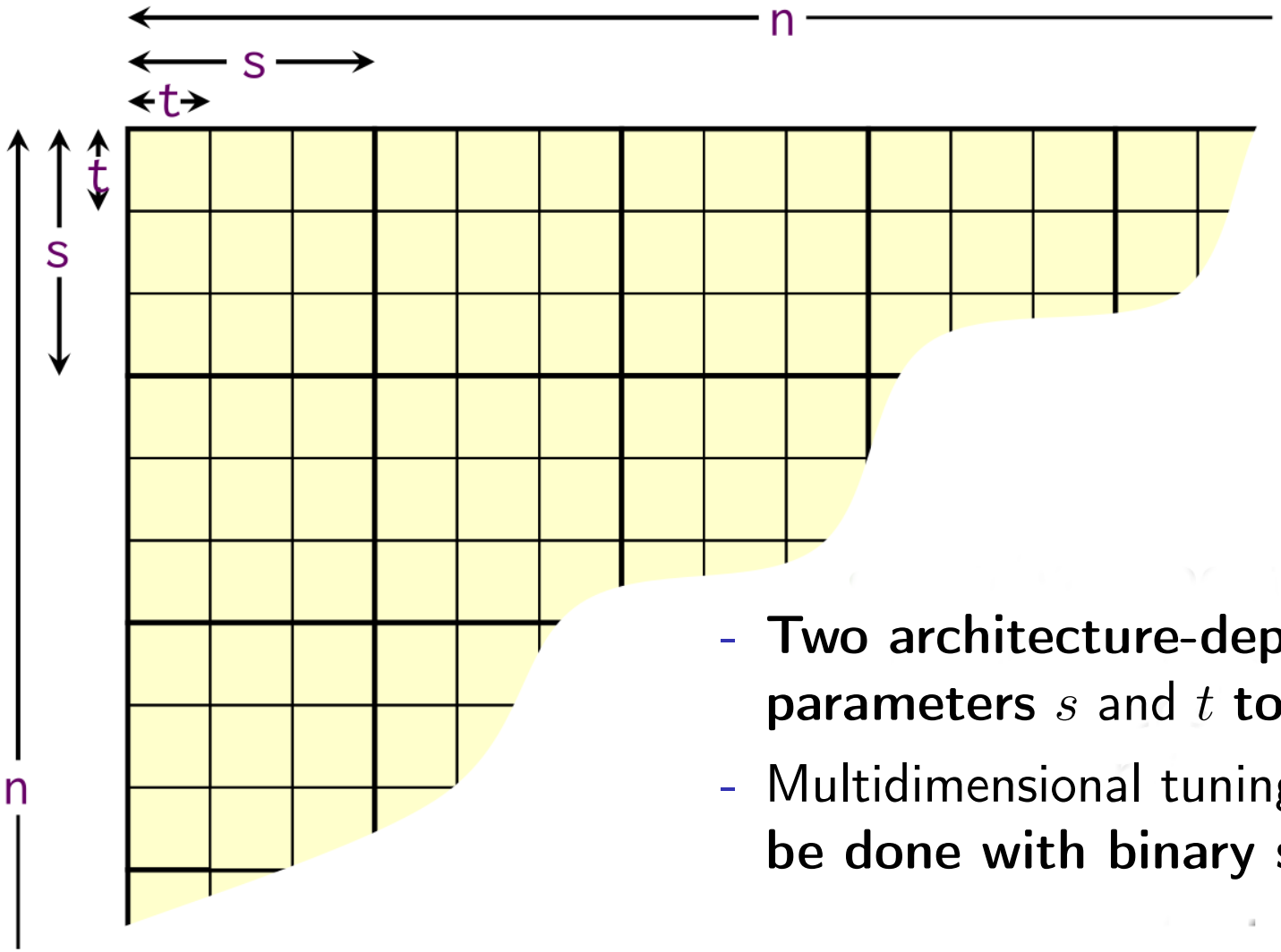
$$Q(n) = \Theta((n/s)^3(s^2/B)) = \Theta(n^3/(BM^{1/2})) \rightarrow \text{ provably } \textbf{optimal}.$$

# Two-level cache



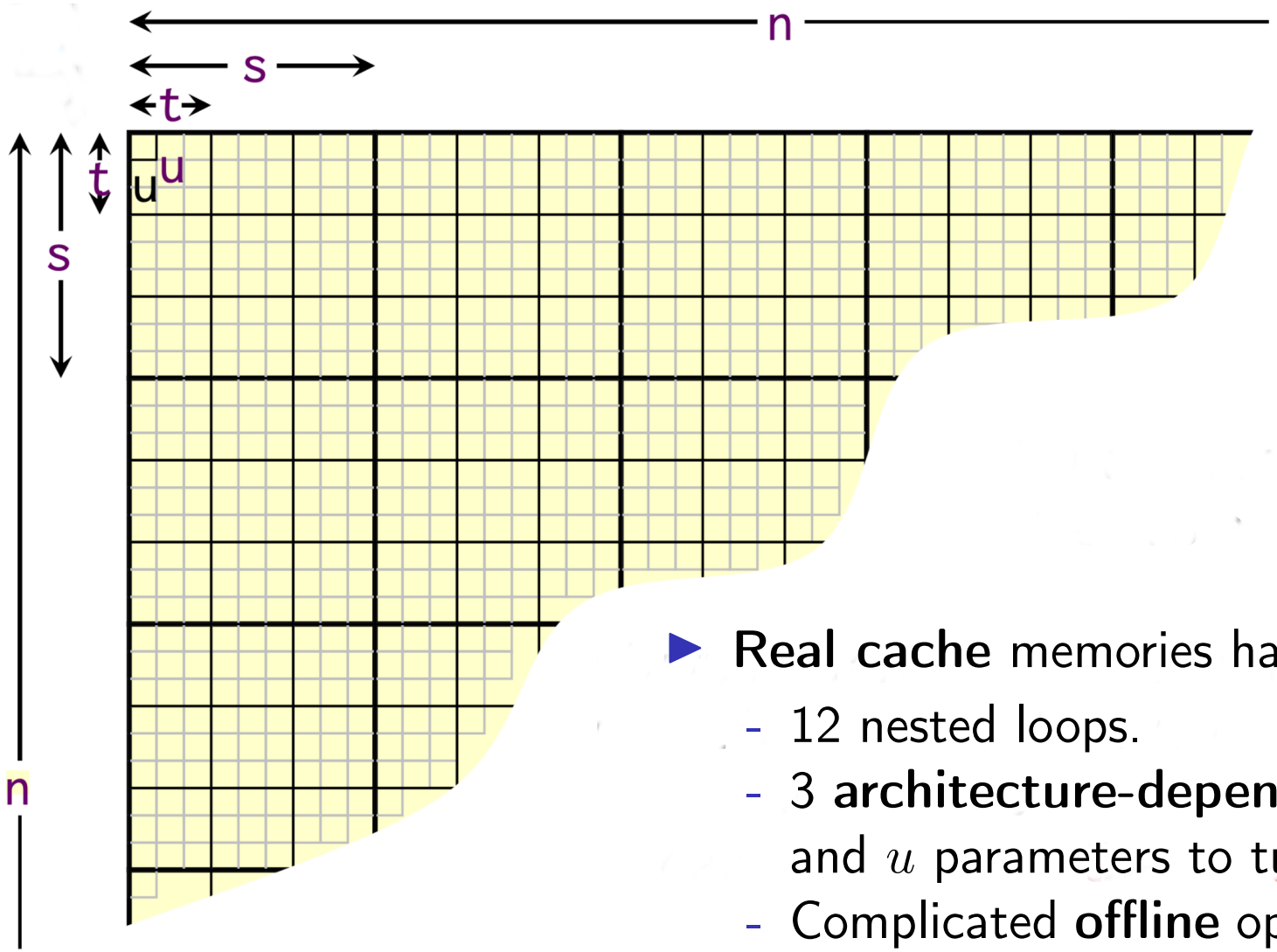

```
for i_2 = 0, s, ..., n − s                                  //zero-based indexing
  for j_2 = 0, s, ..., n − s
    for k_2 = 0, s, ..., n − s
      for i_1 = i_2, i_2 + t, ..., i_2 + s − t
        for j_1 = j_2, j_2 + t, ..., j_2 + s − t
          for k_1 = k_2, k_2 + t, ..., k_2 + s − t
            for i = i_1, ..., i_1 + t − 1
              for j = j_1, ..., j_1 + t − 1
                C[i * n + j] := 0
                  for k = k_1, ..., k_1 + t − 1
                    C[i * n + j] := C[i * n + j]
                                  + A[i * n + k] * B[k * n + j]
```

# Two-level cache, cont'd




- **Two architecture-dependent parameters** $s$ and $t$ **to tune**.
- Multidimensional tuning **cannot be done with binary search**.

# Three-level cache




- **Real cache** memories have 3 levels:
  - 12 nested loops.
  - 3 **architecture-dependent** $s$, $t$ and $u$ parameters to tune.
  - Complicated **offline** optimization.
- Existing implementations:
  - OpenBLAS, BLIS, ATLAS.

# Recursive matrix multiplication

- Divide-and-conquer on $n \times n$ matrices:

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11}B_{11} & A_{11}B_{12} \\ A_{21}B_{11} & A_{21}B_{12} \end{bmatrix} + \begin{bmatrix} A_{12}B_{21} & A_{12}B_{22} \\ A_{22}B_{21} & A_{22}B_{22} \end{bmatrix}$$

Recurrence of 8 multiply-and-add of $n/2 \times n/2$ matrices.

There are two cases of memory access:

- Case 1: $n < cM^{1/2} \implies Q(n) = \Theta(n^2/B)$
- Case 2: otherwise $\implies Q(n) = 8Q(n/2) + \Theta(1)$

  (Master theorem) $\implies Q(n) = \Theta(n^3/(BM^{1/2}))$

# Recursive implementation

- Assuming $n$ is a power of 2:

$\texttt{recursive_gemm}(A, B, C, n):$ $\quad //C_{1:n,1:n} = 0$

if $(n = 1):$

$c_{11} := c_{11} + a_{11} * b_{11}$ $\quad$ //one-based indexing

else :

$\texttt{recursive_gemm}(A_{1:\frac{n}{2},1:\frac{n}{2}}, B_{1:\frac{n}{2},1:\frac{n}{2}}, C_{1:\frac{n}{2},1:\frac{n}{2}}, n/2)$

$\texttt{recursive_gemm}(A_{1:\frac{n}{2},\frac{n}{2}+1:n}, B_{\frac{n}{2}+1:n,1:\frac{n}{2}}, C_{1:\frac{n}{2},\frac{n}{2}+1:n}, n/2)$

$\texttt{recursive_gemm}(A_{1:\frac{n}{2},1:\frac{n}{2}}, B_{1:\frac{n}{2},\frac{n}{2}+1:n}, C_{1:\frac{n}{2},\frac{n}{2}+1:n}, n/2)$

$\texttt{recursive_gemm}(A_{1:\frac{n}{2},\frac{n}{2}+1:n}, B_{\frac{n}{2}+1:n,\frac{n}{2}+1:n}, C_{1:\frac{n}{2},\frac{n}{2}+1:n}, n/2)$

$\texttt{recursive_gemm}(A_{\frac{n}{2}+1:n,1:\frac{n}{2}}, B_{1:\frac{n}{2},1:\frac{n}{2}}, C_{\frac{n}{2}+1:n,1:\frac{n}{2}}, n/2)$

$\texttt{recursive_gemm}(A_{\frac{n}{2}+1:n,\frac{n}{2}+1:n}, B_{\frac{n}{2}+1:n,1:\frac{n}{2}}, C_{\frac{n}{2}+1:n,1:\frac{n}{2}}, n/2)$

$\texttt{recursive_gemm}(A_{\frac{n}{2}+1:n,1:\frac{n}{2}}, B_{1:\frac{n}{2}:n,\frac{n}{2}+1:n}, C_{\frac{n}{2}+1:n,\frac{n}{2}+1:n}, n/2)$

$\texttt{recursive_gemm}(A_{\frac{n}{2}+1:n,\frac{n}{2}+1:n}, B_{\frac{n}{2}+1:n,\frac{n}{2}+1:n}, C_{\frac{n}{2}+1:n,\frac{n}{2}+1:n}, n/2)$

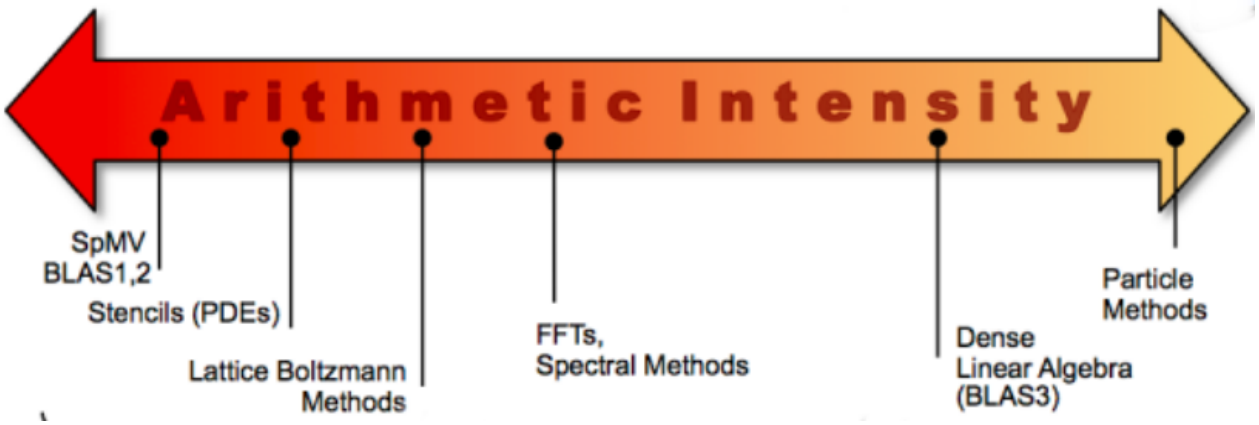
# Efficient cache-oblivious algorithms

- **Cache complexity** of `recursive_gemm` **similar to** that of **tiled matrix multiply**.
- Implementation of `recursive_gemm` **not dependent on memory layout**. It is a **cache-oblivious** algorithm.
- The best cache-oblivious algorithm work on arbitrary rectangular matrices and perform binary splitting (instead of 8-way).
- Recursive matrix-multiply is **typically not implemented in BLAS**, **despite good theoretical cache behavior**, because:
  - It incurs overhead from recursion and function calls.
  - Difficult to efficiently vectorize and parallelize on modern hardware.
  - Hard to outperform hand-optimized, architecture-aware, cache-aware implementations.
- Recursive matrix multiply is more used as a pedagogical tool for illustrating cache efficiency and divide-and-conquer design.
- Recursion is the basis of other fast matrix multiply algorithms, e.g., Strassen's method.

# Overview and principles of communication avoidance

# Reducing data movement

- Principles to reduce communication:
  - **Maximize arithmetic intensity**: Perform more floating-point operations per byte of data moved.




  - **Exploit data locality**: Reuse data in fast memory (cache/registers) as much as possible.
  - **Aggregate communication**: Communicate in bulk instead of fine-grained messages.
- Associated challenges:
  - **Numerical stability**: Communication-avoiding variants may be less stable or harder to analyze.
  - **Architecture-specific tuning**: Optimal strategies depend on cache hierarchy, network topology, etc...

# Sample speedups

- Demmel and Grigori (2016) documents the following speedups achieved by reformulating NLA and other kernels to reduce communication:
  - Up to **12X** faster for **2.5D matrix multiply** on 64K core IBM BG/P
  - Up to **3X** faster for **tensor contractions** on 2K core Cray XE/6
  - Up to **6.2X** faster for **All-Pairs-Shortest-Path** on 24K core Cray CE6
  - Up to **2.1X** faster for **2.5D LU** on 64K core IBM BG/P
  - Up to **11.8X** faster for **direct N-body** on 32K core IBM BG/P
  - Up to **13X** faster for **Tall Skinny QR** on Tesla C2050 Fermi NVIDIA GPU
  - Up to **6.7X** faster for **symeig(band A)** on 10 core Intel Westmere
  - Up to **2X** faster for **2.5D Strassen** on 38K core Cray XT4
  - Up to **4.2X** faster for MiniGMG benchmark bottom solver, using **CA-BiCGStab** (**2.5X** for overall solve)
  - Up to **2.5X** for combustion simulation code

Demmel J. & Grigori L. (2016) Introduction to communication-avoiding algorithms. CS294/Math270, UC Berkeley.

# Sparse matrix-vector product (SpMV)

# Low data locality of SpMV

- Recall the CSR data structure:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & a_{43} & 0 \end{bmatrix}$$

$$\texttt{val} = [a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{33}, a_{34}, a_{43}]$$

$$\texttt{col_idx} = [1, 2, 3, 1, 2, 3, 4, 3]$$

$$\texttt{row_start} = [1, 4, 6, 8, 9]$$

with the following SpMV kernel:

```
for i = 1, ..., n                                  //one-based indexing
  sum := 0
  for j = row_start[i], ..., row_start[i + 1] − 1
    sum := sum + val[j] ∗ x[col_idx[j]]
  y[i] := sum
```

# Low data locality of SpMV, cont'd

- SpMV kernel for CSR data structures:

```
for i = 1, ..., n                                  //one-based indexing
  sum := 0
  for j = row_start[i], ..., row_start[i + 1] − 1
    sum := sum + val[j] ∗ x[col_idx[j]]
  y[i] := sum
```

- **Irregular access** to the components of $\mathtt{x}$:
  - **No spatial locality**, **no temporal locality**.
  - **Every component** of $\mathtt{x}$ loaded for a single multiply-and-add is **trashed immediatly from register**, i.e., no loop unrolling, no SIMD, ...
  - **Sparsity induces lower arithmetic intensity** (AI) than GEMV.
  - **SpMV kernels** are **memory-bound**, even if $\mathtt{x}$ fits in cache.
  - **Performance** of SpMV kernels depends on **data structure**, **non-zero structure** and **hardware**.

# Register blocking

- Improve data locality by promoting **register reuse**:
  - **Register reuse** is **achieved** for the source vector **by blocking**, also known as **tiling**, i.e., relying on contiguous storage of non-zero blocks in the matrix.
  - **Registers** are small, i.e., only **store a handfull of data**:

    $\Longrightarrow$ small $r \times c$ block sizes : $r$ values of destination vector + $c$ values of source vector + $rc$ values of a block are stored in register.
  - Blocking allows for **compiler optimizations**, e.g., loop unrolling, pipelining, ...
  - **Optimal block size dependent** on **sparsity strucutre** and **machine architecture**.
  - Considerable **overhead** in changing data structure for register blocking.
  - Applications such as finite element methods (FEM) naturally store small dense blocks in sparse matrices.
  - Most matrices do not have uniform block structures:

    $\Longrightarrow$ **some zero values** are **stored** for non-fully dense blocks.

# Blocked SpMV

- Recall the BSR (or BCSR) data structure format:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & a_{43} & 0 \end{bmatrix}$$

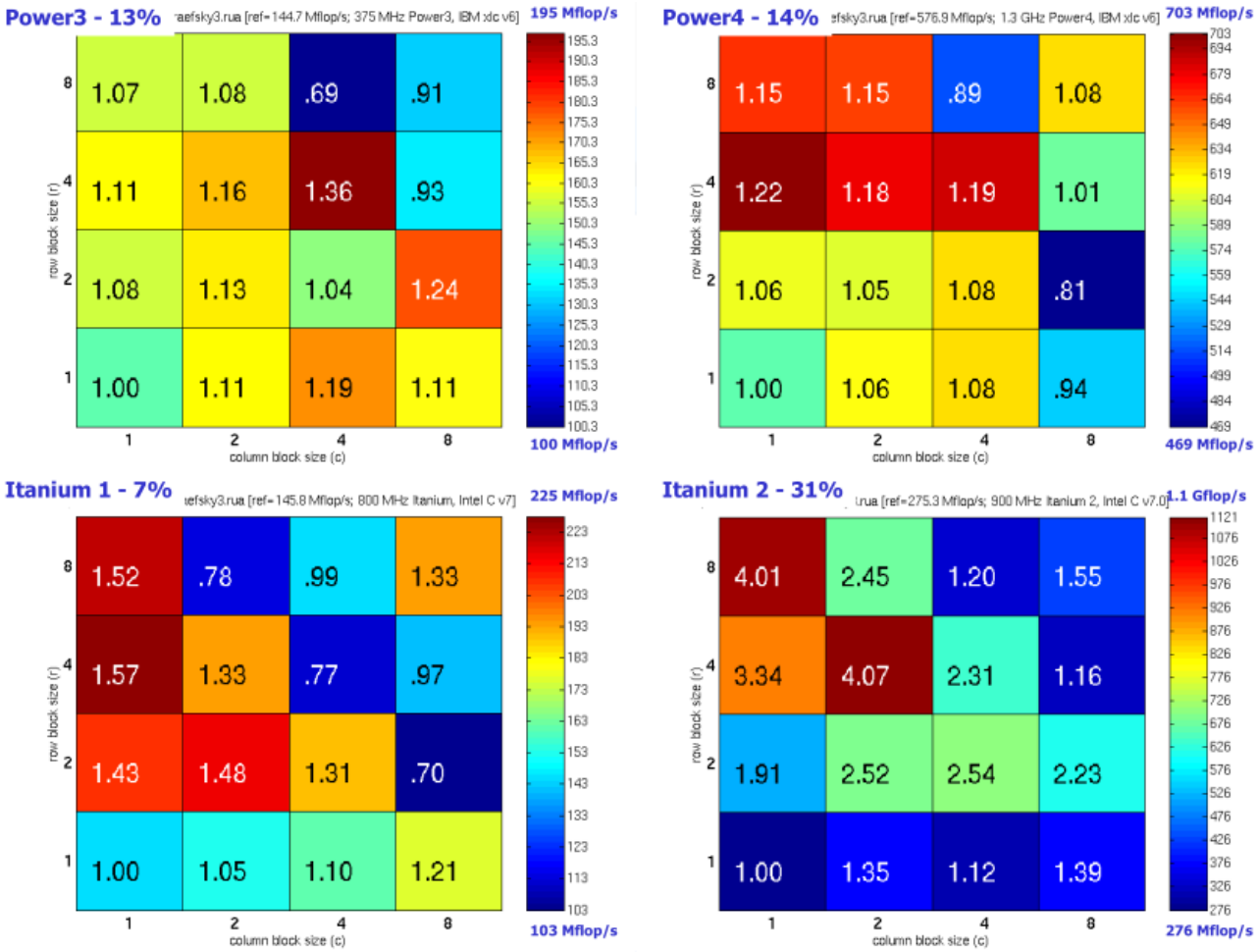
$$\begin{aligned} \texttt{r} = 2,\ \texttt{c} = 2 \\ \texttt{val} = [a_{11}, a_{12}, a_{21}, a_{22}, a_{13}, 0, 0, 0, \\ a_{33}, a_{34}, a_{43}, 0] \\ \texttt{col_idx} = [1, 2, 2],\ \texttt{row_start} = [1, 3, 4] \end{aligned}$$

with the following SpMV kernel:

```
for i = 1, ..., n/r                                   //one-based indexing
  for j = row_start[i], ..., row_start[i+1] - 1
    for ii = 1, ..., r
      sum := 0
      for jj = 1, ..., c
        sum := sum +
         val[(j-1)*rc + (ii-1)*c + jj] * x[(col_idx[j] - 1)*rc + jj]
      y[(i-1)*r + ii] := sum
```

# Tuning of register blocks

- Optimal block size for register reuse depends on **sparsity** and **machine architecture**:
  - Automated tuning interface by Vuduc et al. (2005): OSKI




Vuduc, R., Demmel, J. W., & Yelick, K. A. (2005). OSKI: A library of automatically tuned sparse matrix kernels. In Journal of Physics: Conference Series (Vol. 16, No. 1, p. 521). IOP Publishing.

# Encoding block sparsity

- **Zeros stored** in non-fully dense blocks **result in** extra floating-point operations, and more importantly, **extra data movement**.
- Only non-zero values need be stored and fetched when **local block sparsity** is **encoded**:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & a_{43} & 0 \end{bmatrix} \qquad \begin{aligned} \texttt{r} = 2, \texttt{c} = 2 \\ \texttt{val} &= [a_{11}, a_{12}, a_{21}, a_{22}, a_{13}, a_{33}, a_{34}, a_{43}] \\ \texttt{col_idx} = [1, 2, 2],\ \texttt{row_start} &= [1, 3, 4] \\ \texttt{b_map} &= [15, 1, 7] \end{aligned}$$

E.g., $\begin{bmatrix} a_{33} & a_{34} \\ a_{43} & 0 \end{bmatrix} \implies$ sparsity pattern $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \implies$ unrolled to $1110$

$$\implies 1 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 = 7.$$

- **if statements** in SpMV's loop **decrease efficiency**. Two approaches: operate on zeros (Buluç et al., 2011), de Bruijn sequences (Kannan, 2013).

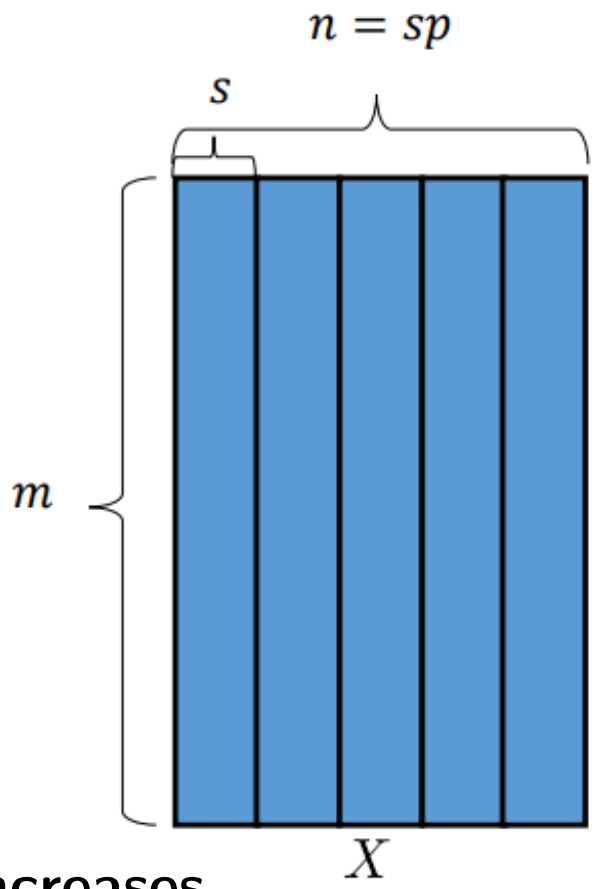
Buluç, A., Williams, S., Oliker, L., & Demmel, J. (2011). Reduced-bandwidth multithreaded algorithms for sparse matrix-vector multiplication. In 2011 IEEE International Parallel & Distributed Processing Symposium (pp. 721-733)

Kannan, R. (2013). Efficient sparse matrix multiple-vector multiplication using a bitmapped format. In 20th Annual International Conference on High Performance Computing (pp. 286-294).

# Block Gram-Schmidt procedures

# Block Gram-Schmidt procedures

- **Gram-Schmidt (GS)** procedure:
  - Returns the QR factorization of an $m \times n$ matrix $X$ by **orthogonalizing vectors**, **one-at-a-time**, against previously orthnogonalized vectors.
- **Block Gram-Schmidt (BGS)** procedure:
  - Works on **blocks of $s$ vectors** at a time.
  - Procedure fully specified upon definition of a **projector $\Pi^{(i)}$ onto** range$(Q_{:,1:is})^{\perp}$ and an **IntraOrtho** orthogonalization procedure:




IntraOrtho$(X_{:,1:s}) \mapsto (Q_{:,1:s}, R_{1:s,1:s})$

for $i = 2, \ldots, p$ //one-based indexing

$Q_{:,(i-1)s+1:is} := \Pi^{(i-1)} X_{:,(i-1)s+1:is}$

IntraOrtho$(Q_{:,(i-1)s+1:is}) \mapsto$

$(Q_{:,(i-1)s+1:is}, R_{(i-1)s+1:is,(i-1)s+1:is})$

  - Applying $\Pi^{(i)}$ to blocks of $s$ vectors at a time **increases arithmetic intensity**, and **decreases synchronizations**.

# Block classical Gram-Schmidt (BCGS)

- BCGS is formed with $\Pi^{(i)} := I_n - Q_{:,1:is}Q_{:,1:is}^T$ which leads to

| | |
|---|---|
| IntraOrtho$(X_{:,1:s}) \mapsto (Q_{:,1:s}, R_{1:s,1:s})$ | //one-based indexing |
| for $i = 2, \ldots, p$ | |
| $R_{1:(i-1)s,(i-1)s+1:is} := Q_{:,1:(i-1)s}^T X_{:,(i-1)s+1:is}$ | //BLAS-3 |
| $W := X_{:,(i-1)s+1:is} - Q_{:,1:(i-1)s}R_{1:(i-1)s,(i-1)s+1:is}$ | //BLAS-3 |
| IntraOrtho$(W) \mapsto (Q_{:,(i-1)s+1:is}, R_{(i-1)s+1:is,(i-1)s+1:is})$ | |

**All $s$ vectors** of the block treated by IntraOrtho are **simultaneaously available**.

Therefore, **IntraOrtho needs not** necessarily **be Gram-Schmidt**.

The QR factorization can be computed by Cholesky QR (CholQR), tall and skinny QR (TSQR), Householder QR (HouseQR), ...:

$$\text{IntraOrtho} \in \{\text{CGS}, \text{CGS2}, \text{MGS}, \text{CholQR}, \text{CholQR2}, \text{TSQR}, \text{HouseQR}, \ldots\}$$

Each so-specified variant is referred to as BCSG $\circ$ IntraOrtho.

# Block classical Gram-Schmidt reorthogonalized (BCGS2)

- BCGS is stabilized by reorthogonalizing either by running BCSG twice, or by repeating each inner loop (BCGS2):

$\text{IntraOrtho}(X_{:,1:s}) \mapsto (Q_{:,1:s}, R_{1:s,1:s})$ //one-based indexing

for $i = 2, \ldots, p$

$R^{(1)}_{1:(i-1)s,(i-1)s+1:is} := Q^T_{:,1:(i-1)s} X_{:,(i-1)s+1:is}$ //BLAS-3

$W := X_{:,(i-1)s+1:is} - Q_{:,1:(i-1)s} R^{(1)}_{1:(i-1)s,(i-1)s+1:is}$ //BLAS-3

$\text{IntraOrtho}(W) \mapsto (\widehat{Q}, R^{(1)}_{(i-1)s+1:is,(i-1)s+1:is})$

$R^{(2)}_{1:(i-1)s,(i-1)s+1:is} := Q^T_{:,1:(i-1)s} \widehat{Q}$ //BLAS-3

$W := \widehat{Q} - Q_{:,1:(i-1)s} R^{(2)}_{1:(i-1)s,(i-1)s+1:is}$ //BLAS-3

$\text{IntraOrtho}(W) \mapsto (Q_{:,(i-1)s+1:is}, R^{(2)}_{(i-1)s+1:is,(i-1)s+1:is})$

$R_{1:(i-1)s,(i-1)s+1:is} := R^{(1)}_{1:(i-1)s,(i-1)s+1:is} + R^{(2)}_{1:(i-1)s,(i-1)s+1:is} R^{(1)}_{(i-1)s+1:is,(i-1)s+1:is}$

$R_{(i-1)s+1:is,(i-1)s+1:is} = R^{(2)}_{(i-1)s+1:is,(i-1)s+1:is} R^{(1)}_{(i-1)s+1:is,(i-1)s+1:is}$

We refer to associated variants as BCGS2 $\circ$ IntraOrtho where

IntraOrtho $\in \{$CGS, CGS2, MGS, CholQR, CholQR2, TSQR, HouseQR, $\ldots\}$

# Block modified Gram-Schmidt (BMGS)

- BMGS is formed with

$$\Pi^{(i)} := (I_n - Q_{:,(i-1)s+1:is} Q^T_{:,(i-1)s+1:is}) \dots (I_n - Q_{:,1:s} Q^T_{:,1:s})$$

which leads to

IntraOrtho$(X_{:,1:s}) \mapsto (Q_{:,1:s}, R_{1:s,1:s})$ //one-based indexing

for $i = 2, \dots, p$

  $W := X_{:,(i-1)s+1:is}$

  for $j = 1, \dots, i-1$

    $R_{(j-1)s+1:js,(i-1)s+1:is} := Q^T_{:,(j-1)s+1:js} W$ //BLAS-3

    $W := W - Q_{:,(j-1)s+1:js} R_{(j-1)s+1:js,(i-1)s+1:is}$

  IntraOrtho$(W) \mapsto (Q_{:,(i-1)s+1:is}, R_{(i-1)s+1:is,(i-1)s+1:is})$

We refer to associated variants as BMGS $\circ$ IntraOrtho where

IntraOrtho $\in \{$CGS, CGS2, MGS, CholQR, CholQR2, TSQR, HouseQR, $\dots\}$

# Block randomized Gram-Schmidt (BRGS)

- BRGS is formed by with $\Pi^{(i)} := I_n - Q_{:,1:is}(\Theta Q_{:,1:is})^\dagger \Theta$ which leads to

Compute $Q_{:,1:s}$ and $R_{1:s,1:s}$ //RGS or RCholQR

s.t. $\Theta Q_{:,1:s} R_{1:s,1:s} = \Theta X_{:,1:s}$

and $(\Theta Q_{:,1:s})^T \Theta Q_{:,1:s} = I_s$

$S_{:,1:s} := \Theta Q_{:,1:s}$ //one-based indexing

for $i = 2, \ldots, p$

$P := \Theta X_{:,(i-1)s+1:is}$

$R_{1:(i-1)s,(i-1)s+1:is} := \arg \min_{Y \in \mathbb{R}^{(i-1)s \times s}} \|S_{:,1:(i-1)s} Y - P\|_F$

$Q_{:,(i-1)s+1:is} := X_{:,(i-1)s+1:is} - Q_{:,1:(i-1)s} R_{1:(i-1)s,(i-1)s+1:is}$

Compute $Q_{:,(i-1)s+1:is}$ and $R_{(i-1)s+1:is,(i-1)s+1:is}$ //RGS or RCholQR

s.t. $\Theta Q_{:,(i-1)s+1:is} R_{(i-1)s+1:is,(i-1)s+1:is} = \Theta X_{:,(i-1)s+1:is}$

and $(\Theta Q_{:,(i-1)s+1:is})^T \Theta Q_{:,(i-1)s+1:is} = I_s$

$S_{:,(i-1)s+1:is} := \Theta Q_{:,(i-1)s+1:is}$

# Stability aspects

- The loss of orthogonality $\|I_n - Q^T Q\|_2$ achieved in finite precision by BGS algorithms depends on $\Pi^{(i)}$ as well as on IntraOrtho.

  If IntraOrtho is unconditionally stable, i.e., if IntraOrtho$(X) \mapsto (Q, R)$ is such that $\|I_n - Q^T Q\|_2 = \mathcal{O}(\varepsilon)$ irrespective of $\kappa(X)$, then:

| BGS | $\|I_n - Q^T Q\|_2$ | Assumption on $\kappa(X)$ | Reference(s) |
|---|---|---|---|
| BCGS | $\mathcal{O}(\varepsilon)\kappa^{n-1}(X)$ | $\mathcal{O}(\varepsilon)\kappa(X) < 1$ | conjecture |
| BCGS2 | $\mathcal{O}(\varepsilon)$ | $\mathcal{O}(\varepsilon)\kappa(X) < 1$ | Barlow and Smoktunowicz (2013) |
| BMGS | $\mathcal{O}(\varepsilon)\kappa(X)$ | $\mathcal{O}(\varepsilon)\kappa(X) < 1$ | Jalby and Philippe (1991) |

- Most orthogonalization procedures are not unconditionally stable:

| IntraOrtho | $\|I_n - Q^T Q\|_2$ | Assumption on $\kappa(X)$ | Reference(s) |
|---|---|---|---|
| CGS | $\mathcal{O}(\varepsilon)\kappa^{n-1}(X)$ | $\mathcal{O}(\varepsilon)\kappa(X) < 1$ | Kiełbasiński (1974) |
| CGS2 | $\mathcal{O}(\varepsilon)$ | $\mathcal{O}(\varepsilon)\kappa(X) < 1$ | Giraud et al. (2005) |
| MGS | $\mathcal{O}(\varepsilon)\kappa(X)$ | $\mathcal{O}(\varepsilon)\kappa(X) < 1$ | Björck (1967) |
| CholQR | $\mathcal{O}(\varepsilon)\kappa^2(X)$ | $\mathcal{O}(\varepsilon)\kappa^2(X) < 1$ | Yamamoto et al. (2015) |
| CholQR2 | $\mathcal{O}(\varepsilon)$ | $\mathcal{O}(\varepsilon)\kappa^2(X) < 1$ | Yamamoto et al. (2015) |
| HouseQR | $\mathcal{O}(\varepsilon)$ | none | Section 10 in Higham (2002) |
| TSQR | $\mathcal{O}(\varepsilon)$ | none | Mori et al. (2012) |

# Stability aspects, cont'd

- Other results available:
  Jalby and Philippe (1991):
  - BMGS $\circ$ MGS behaves like CGS.
  - BMGS $\circ$ MGS2 is as stable as MGS.

  Carson et al. (2022):
  - BMGS $\circ$ any IntraOrtho with $\|I_n - Q^T Q\| \leq \mathcal{O}(\varepsilon)$ is as stable MGS.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# $s$-step iterative solvers

# Matrix power kernels

# Homework problems

# Practice session